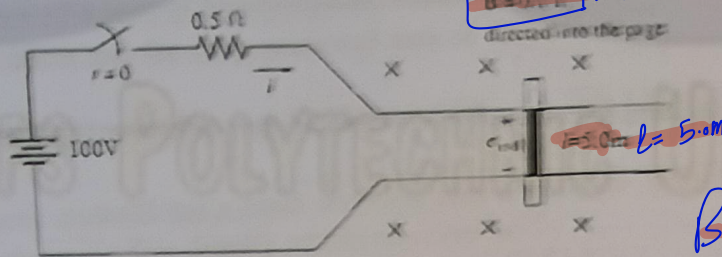




جامعة الملك سعود

$B = 0.1 \text{ T}$ directed into the page



Inhalt:

تحصیل

$$B = 0.1 \text{ ng @ v line}$$

وبالاستبانة كانت 0.01

1. If a 20 N force is opposing the motion of the machine, the steady-state current consumed by the machine is:
- a) 1.0 A b) 4.0 A c) 40 A d) 200 A
2. 27) If a 20 N force is opposing the motion of the machine, the steady-state velocity of the machine is:
- a) 160.0 m/s b) 120.0 m/s c) 100 m/s d) 16.0 m/s

XX

3. A single phase transformer has 110 turns on the secondary and 1000 turns on primary. If the transformer is connected to 110 volt supply while 3 kVA load is connected to the secondary winding. Assuming ideal transformer, the load current is: 30 A

- a) 330A **b) 247.9A** c) 272.7A d) None of the previous

برای تهیه رست باغچه

4. A single phase transformer has 110 turns on the secondary and 1000 turns on primary. If the transformer is connected to 110 volt supply while 3 kVA load is connected to the secondary winding. Assuming ideal transformer, the load impedance as referred to the primary is:

- a) 0.4Ω b) 4.0Ω c) 3.96Ω d) 403Ω

(c) [REDACTED] (b)(7)(C), (b)(7)(D)

(d) [REDACTED] (b)(7)(C), (b)(7)(D)

6. A rotor of an electric machine rotates at constant speed and makes 2500 revolutions per minute. The time required for one complete revolution in seconds is:

- a) $\frac{1500}{2\pi}$ b) $\frac{1500}{120\pi}$ c) 0.024 d) 0.04

صحن یغیر الارحام ۲۵۲ اہم رحمت

7. The rated speed of a motor is 1800 RPM. If the rated torque of this motor is 140 Nm, the rated power of this motor is;

- (a) 5hp b) 10hp c) 20hp d) 35hp

8. The magnetic flux through a coil having a single turn is varying according to the relation: $\phi = (25t^3 + 10t^2 + 5t - 5) \text{ Wb}$. The induced voltage between the coil terminals at $t = 1 \text{ s}$ is:

- a) $10V$ b) $25V$ c) $50V$ ~~d) $100V$~~

$$\text{inl} = 20 \quad \checkmark \quad \rightarrow 6t^2 + 20t + 6 \quad |_{t=1}$$

$$75 + 20 + 5 = 100$$

$$= 2.5 \rightarrow N = 25 \frac{d\phi}{dt} = 0.1$$

$$e_{ind} = N \frac{d\phi}{dt}$$

نوع
جواب
نوع
نوع
نوع
نوع

$$\phi = BA \rightarrow B = \frac{\phi}{A}$$

$$e_{ind} = (v \times B) \cdot l$$

Q

14. We laminate transformer core to reduce

a) eddy current loss

b) hysteresis loss

c) both eddy current and hysteresis loss

d) ohmic loss

15. To conduct the open circuit test, test is conducted on the

a) LV side

b) HV side

c) primary

d) secondary

eddy current

LV side
Zigzagging
with
iron

1	2	3	4	5	6	7	8	9	10
c	a	b	a	b	c	a	b	c	d

11	12	13	14	15
b	...	...	...	a

1. The open circuit test and short circuit test are performed on a transformer to determine the:
 - a. Equivalent circuit of transformer
 - b. Voltage regulation of transformer
 - c. Efficiency of transformer
 - d. All the above
 - e. None of the above
2. Which of the following loss varies with the load:
 - a. Copper loss
 - b. Hysteresis loss
 - c. Eddy current loss
 - d. All losses in transformer vary with load
3. On ideal transformer draw current with no load connected to secondary winding:
 - a. False
 - b. True
4. On open circuit transformer test, voltage at rated frequency is applied to:
 - a. HV side
 - b. LV side
 - c. LV and HV side
 - d. LV or HV side
5. Transformer open circuited test, input power consists of core losses in transformer:
 - a. False
 - b. True
6. The primary and secondary winding of transformer have linked each other by:
 - a. Mutual conduction
 - b. Mutual induction
 - c. Not linked at all
 - d. Self-induction

Handwritten signature

30. The magnetic reluctance of a core of ferromagnetic material with relative of 4000, a mean path length of 100cm and the cross-sectional area of 150cm² is:
- a. 0.01667 A.turn/Wb
 - b. 2500 A.turn/Wb
 - c. 8750 A.turn/Wb
 - d. 13263 A.turn/Wb

$$M = \frac{\mu_r \mu_0}{l} \times \frac{1}{A} \times 4\pi \times 10^{-7} \times 100 \times 150 \times 10^{-4}$$

$$l = 1 \text{ m} \quad A = 0.015 \text{ m}^2$$

31. A core of ferromagnetic material has a cross-sectional area of 100cm², if the magnetic flux through this core is 0.01Wb, the magnetic flux density within this core is:
- a. 0.1T
 - b. 100T
 - c. 10T
 - d. 1.0T

$$A = 0.01 \text{ m}^2$$

$$\phi = 0.01 \text{ Wb}$$

$$B = \frac{\phi}{A} = \frac{0.01}{0.01} = 1 \text{ T}$$

ليث عمرو

Electrical Machine
Mid Exam

32. A 3-phase transformer has its primary connected in delta and secondary in star, the secondary to primary turns ratio per phase is 5, What would be the secondary voltage if the primary voltage is 400V:
- a. 80V
 - b. 3364V
 - c. 2000V
 - d. 138V
 - e. None of the above
33. Open circuit test on transformers is conducted so as to get:
- a. Hysteresis losses
 - b. Copper losses
 - c. Core losses
 - d. Eddy current losses

Core = Hysteresis / Eddy

8. Transformer transform:

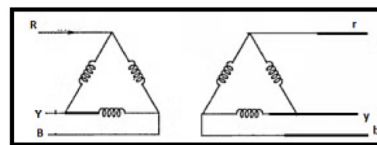
- a. Current
- b. Voltage (Check for this)
- c. Power
- d. Frequency

Power V

9. The property of the material which oppose the creation of magnetic flux it's called:

- a. Reluctivity
- b. Permeability (Check for this)
- c. Magnetomotive force
- d. Reluctance

10. Below is the circuit diagram for delta-delta transformer have a transformation ratio 'k'. The current flowing in the windings of phase r-y in the below diagrams is:



- a. $k^2 i / 1.73$
- b. $i / k \cdot 1.73$
- c. $1.37 \cdot k \cdot i$
- d. $3 \cdot k \cdot i$

11. Secondary current of a step-down transformer is:

- a. Lower than primary current
- b. Higher than primary current
- c. Equal to primary current
- d. Double than primary current

12. The core of transformer must have:

- a. Low reluctance
- b. High reluctance
- c. Low resistance
- d. high resistance

13. An ideal transformer is one which has:

- a. No winding resistance
- b. No leakage reactance
- c. No losses
- d. All of above

ليث عمرو

ليث عمرو

14. Iron loss in transformer can be determined by:

- a. Short circuit test
- b. Open circuit test
- c. Both short circuit and open circuit tests
- d. None of the above

15. The law describing the electromotive force induced in a conductor placed in varying magnetic field is:

- a. Ohm's law
- b. Gauss's law
- c. Faraday's law
- d. Ampere's law

16. The unit of the magnetic flux is:

- a. Henry
- b. Webber
- c. Tesla
- d. Ampere/meter

17. A constant current flow in a horizontal wire in the plane of the paper from east to west as shown in figure. The direction of magnetic field at a point will be north to south:

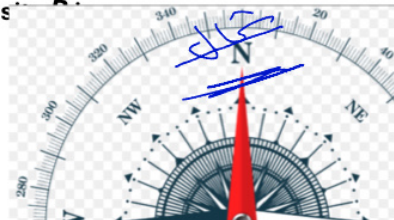
- a. Directly above the wire
- b. Directly below the wire
- c. In the plane of the paper, In the north side of the wire.
- d. In the plane of the paper, In the south side of the wire.

18. The electromotive force induced in a conductor of length l moving at velocity F in a magnetic field having a flux density B is:

- a. $e_{ind} = (F \times B) \times l$
- b. $e_{ind} = (V \cdot B) \cdot l$
- c. $e_{ind} = (V \cdot B) \times l$
- d. $e_{ind} = (V \times B) \cdot l$

19. The force F on a current carrying conductor of length l and current I , placed perpendicular in a uniform magnetic field of flux dens

- a. $F = I/B$
- b. $F = B/I$
- c. $F = BI$
- d. $F = I/BI$



5. A transformer transform

- a. Current
- ☒ b. Voltage
- c. Power
- d. Frequency

Muath
Makawi

21

7. The property of material which oppose the creation of magnetic flux in it is known as

- A Reluctivity
- ☒ C Permeance
- B Magnetomotive force
- D Reluctance

8. Point out the wrong statement. Magnetic leakage is undesirable in electrical machine because it

- A Lower their power efficiency
- ☒ B Increase their cost of manufacture
- C Lead to their increased weight
- D Produce fringing

9. Transformer Parameters can be determined through

- A Faradys law
- ☒ B Open circuit test and short circuit test
- C Y-Δ connection set-up
- D All the above

10. How 3-phase transformers are constructed?

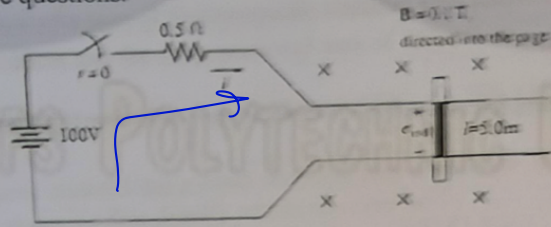
- a) A bank of 3 single phase transformers
- ☒ b) Single 3-phase transformer or a bank of 3 1-phase transformers
- c) A single 3-phase transformer with the primary and secondary of each phase wound on three legs of a common core
- d) By different method

سيز 80

اخر سوال 15 في 100 الاجاب بطلو 10

$\sin \theta = 0 \rightarrow \theta = 180 \rightarrow$ في 180

For Questions 1 and 2, the non-loaded linear DC machine shown in the figure below, choose the correct answer for the following five questions.



~~حالت بالانتقال~~

~~تحت~~

~~$B = 0.1$ in θ in line~~

~~وبالانتقال كانت 0.01~~

1. If a 20N force is opposing the motion of the machine, the steady-state current consumed by the machine is:

- a) 1.0A b) 4.0A c) 40A d) 200A

2. 27) If a 20N force is opposing the motion of the machine, the steady-state velocity of the machine is:

- a) 160.0m/s b) 120.0m/s c) 100m/s d) 16.0m/s

بالقوة الـ 20N التي هي ~~Motor~~ في اتجاه اليمين

$$-100 + 0.5 I + e_{ind} = 0 \rightarrow \text{لحساب سرعة الحركة}$$

$$F = 20 \rightarrow \text{علاقة القوة بالتيار}$$

$$F = I l B \rightarrow \text{معطى}$$

$$20 = I \times 5 \times 0.1$$

$$I = 40 \text{ A} \rightarrow \text{التيار المطلوب}$$

ملاحظة: بالانتقال يعني كان طالب 3 افرد

~~I , e_{ind} , V_{ss}~~ وكانت $B = 0.01$

~~فركزوا بطريقة حل (من حفظ)~~

~~V_{starested}~~ ~~الفرع الثاني~~

eing = $V_{ss} BL$

~~في حرفه~~

والفرع الثاني I

$$-100 + 0.5 I + \text{eing} = 0$$

$$-100 + 0.5 \times 40 + \text{eing} = 0$$

$$\text{eing} = 80$$

$$\text{eing} = V_{ss} BL$$

$$80 = V_{ss} \times 0.1 \times 5$$

→ 25/1.25

$$\cancel{V_{ss}} = \frac{80}{0.1 \times 5} = \underline{\underline{160}}$$

3. A single phase transformer has 110 turns on the secondary and 1000 turns on primary. If the transformer is connected to 110 volt supply while 3 kVA load is connected to the secondary winding. Assuming ideal transformer, the load current is: ~~330A~~ ~~272.7A~~ ~~403Ω~~
- a) 330A b) 247.9A c) 272.7A d) None of the previous
4. A single phase transformer has 110 turns on the secondary and 1000 turns on primary. If the transformer is connected to 110 volt supply while 3 kVA load is connected to the secondary winding. Assuming ideal transformer, the load impedance as referred to the primary is:
- a) 0.4Ω b) 4.0Ω c) 3.96Ω d) 403Ω

$$\frac{N_P}{N_S} = a = \frac{V_P}{V_S} = \frac{I_S}{I_P}$$

$$\frac{1000}{110} = a \quad a s \quad \frac{100}{11} = \frac{110}{V_S}$$

$$100V_S = 1210$$

$$V_S = 12.1 \text{ V}$$

$$S = 3k = I_S V_S$$

$$I_S = \frac{3k}{12.1} = 247.9 \text{ A}$$

$$R_l \rightarrow \frac{12.1}{247.9} = 0.04 \rightarrow \text{in secondary}$$

$$\text{in primary} = 22 a^2 = 0.04 \times \left(\frac{100}{11}\right)^2$$

$$= 4.03$$

Q3) Solve and show your work

$$a = 10$$

(7 Marks)

A single phase transformer has 100 turns on the secondary and 1000 turns on primary. If transformer is connected to 110 volt supply while 3 kVA load connected to the secondary winding. Assuming ideal transformer, find

- A load impedance
- B load voltage
- C load impedance as referred to the primary

$$V_P = 110$$

$$\frac{N_P}{N_S} = 10 = a = \frac{V_P}{V_S}$$

$$10 \times \frac{110}{V_S} =$$

$$V_S = 110$$

$$3 \text{ k} = V_S I_S$$

$$\text{So } I_S = \frac{3000}{11}$$

$$= 272.72 \text{ A}$$

$$\text{load impedance} = \frac{V_S}{I_S} = \frac{11}{272.72} = 0.04$$

$$a^2 Z_L = 100 \times 0.04 = 4 \text{ } \Omega$$

RPM

6. A rotor of an electric machine rotates at constant speed and makes 2500 revolutions per minute. The time required for one complete revolution in seconds is:

a) $\frac{1500}{2\pi}$ b) $\frac{1500}{120\pi}$ c) 0.024 d) 0.04

7. The rated speed of a motor is 1800RPM. If the rated torque of this motor is 140Nm, the rated power of this motor is:

a) 5hp b) 10hp c) 20hp d) 35hp

يمكن تغير الارتفاع في سرعة الدوران

$\omega_m \rightarrow$ angular velocity rad/s
 $f_m \rightarrow$ " " " " BPS
 $n_m \rightarrow$ " " " " RPM

سرعة زاوية

دوران

$$n_m = 60 f_m$$

$$f_m = \frac{\omega_m}{2\pi} \rightarrow \text{rad/s}$$

$$\left. \begin{aligned} \omega &= \frac{d\theta}{dt} \\ a &= \frac{d\omega}{dt} \\ \tau &= r \times F \\ P &= \omega \tau \end{aligned} \right\} \text{دوران}$$

RPS or RPM as you like

$$n_m = 60 f_m$$

$$f_m = \frac{n_m}{60} = \frac{2500}{60} = 41.6$$

تسريع زاوي

$$\frac{1}{41.6} = 0.024$$

$$t_s = \frac{1}{f_m}$$

زمن

$$1800 = \text{RPM}$$

7. The rated speed of a motor is 1800RPM. If the rated torque of this motor is 140Nm, the rated power of this motor is:
 a) 5hp b) 10hp c) 20hp d) 35hp

ليكون السرعة ω بالـ rad/s

السرعة بالـ rad/s

والسرعة ω بالـ rad/s

السرعة بالـ rad/s

في القانون قوة

$$W_{ms} = 2\pi f m = W_{ms} = 2\pi \times \text{nm}$$

$$f_m = \frac{1m}{60}$$

60

$$W_m = 188.49$$

Power

16 g

$$P = \cancel{1} W_m$$

$$= \cancel{140} \times 188.49$$

see

$$= 26.38 \times 10^3 \text{ W}$$

hp is 1000 W

$$1 \text{ hp} \rightarrow 746 \text{ W}$$

$$X \rightarrow 26.38 \times 10^3$$

$$= \frac{26.38 \times 10^3}{746} = \underline{\underline{35 \text{ hp}}}$$